

Assessment Tool-2



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Course code! CSA1613

Course Name! Data warehouse and Data mining.

Data mining:-

Data mining is the process of discovering useful, interesting, and previously unknown patterns or knowledge from large volumes of data.

Any Two Basic Data mining Tasks:-

① classification

② clustering

Steps in KDD:-

① Data cleaning

② Data integration

③ Data selection

④ Data Transformation

⑤ Data mining

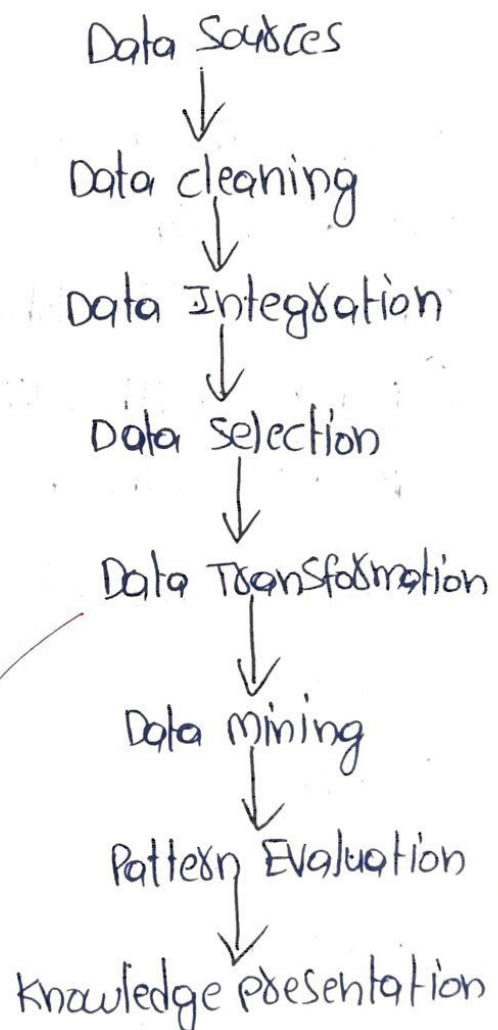
⑥ Pattern Evaluation

⑦ Knowledge presentation

Role of Data preprocessing:-

- * Improves data quality
- * Removes inconsistencies
- * Reduces noise
- * Handles missing values
- * Increases mining accuracy

KDD process Diagram



Integration
multiple

1) Data cleaning

Data cleaning is the process of detecting and correcting inaccurate, incomplete, or inconsistent data.

Two data quality issues:-

① Missing values

② Noisy (incorrect/random) data.

Handling Missing values

methods:-

① Ignore the record.

② Fill manually.

③ Replace with global constant.

④ Replace with attribute mean/median.

⑤ Predict using machine learning.

Handling Noisy Data:-

methods:-

① Binning

② Regression

③ clustering

④ outlier analysis

Data Integration:

Data integration combines data from multiple databases into one consistent dataset.

Challenges:

Data redundancy

Data inconsistency

Handling Redundancy: occurs when same data appears multiple times.

Methods:

* Duplicate detection

* Correlation analysis

* Entity matching

* Schema integration

Handling Inconsistency: occurs when values conflict.

Methods:

* Standardization

* Constraint checking

* Data validation

* Conflict resolution

Example:

Gender: m

Gender: Male

Convert both into one standard format

⑥ Covariance Analysis - covariance analysis

Two variables change together

Positive Covariance $\rightarrow$ Variables

increase together

Negative Covariance $\rightarrow$ one increases while the other

decreases

Zero covariance $\rightarrow$ independent relationship

Two Covariance Formulas

Population Covariance

$$\text{cov}(X, Y) = \frac{\sum (x_i - \mu_x)(y_i - \mu_y)}{N}$$

Sample Covariance

$$\text{cov}(X, Y) = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{n-1}$$

③ Stock Covariance

Data :-

A	B
2	5
3	8
5	10
4	11
6	14

mean:

$$\bar{x} = \frac{2+3+5+4+6}{5} = 4$$

$$\bar{y} = \frac{5+8+10+11+14}{5} = 9.6$$

Products.

$x_i - \bar{x}$	$y_i - \bar{y}$	Product
-2	-4.6	9.2
-1	-1.6	1.6
1	0.4	0.4
0	1.4	0
2	4.4	8.8

Total 20

$$\text{Sample covariance} \cdot \text{cov} = \frac{20}{4} = 5$$

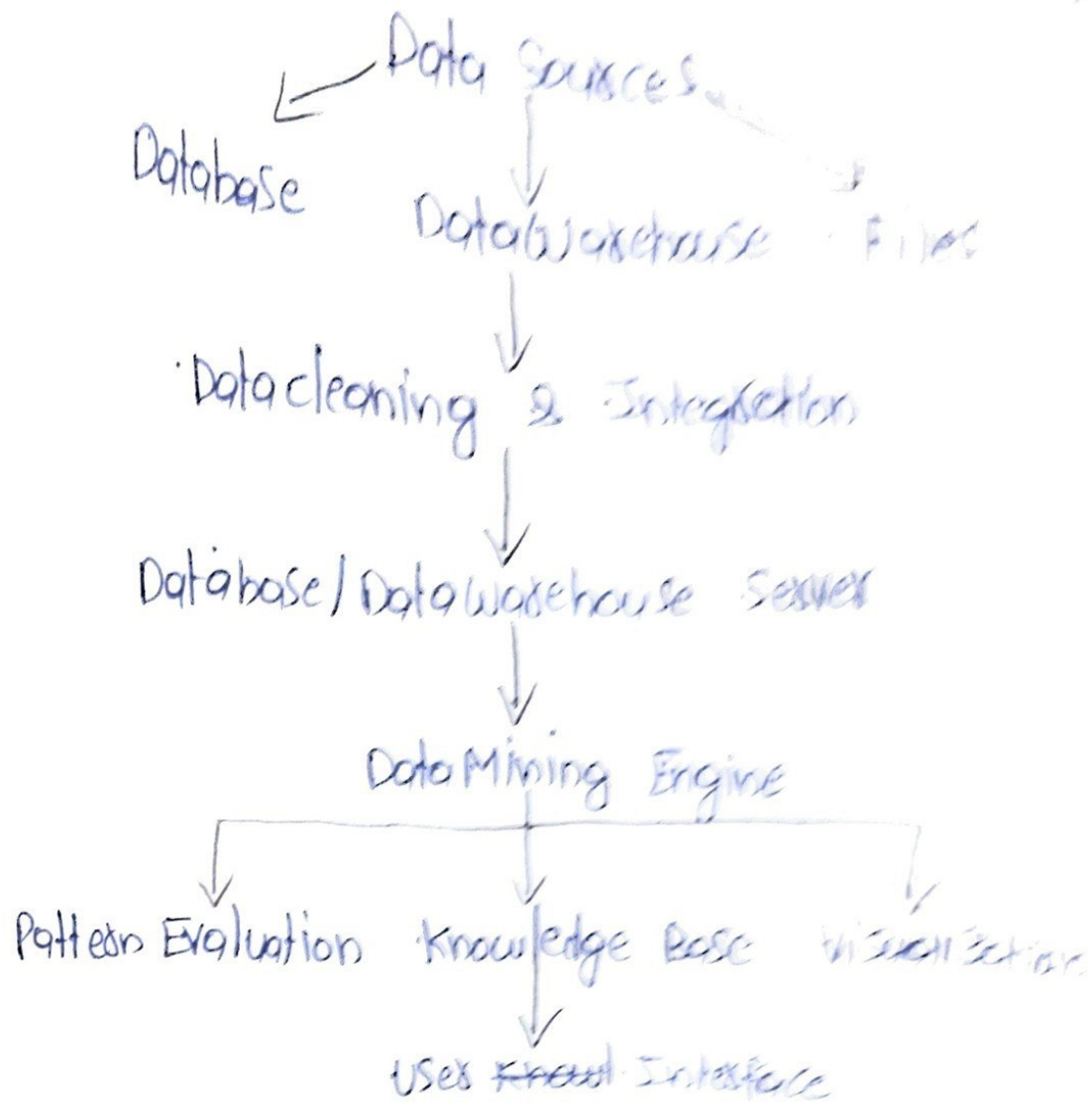
These facts,

* Stock A and stock B rise and fall together.

* They are not independent

Architecture of Data Mining System

Diagram:-



Interaction:-

- * Data flows from sources to preprocessing
- * Clean data enters mining engine
- * Patterns are evaluated
- * Useful knowledge is presented to users

④ Covariance:

Given,

$$X = \{4, 5, 6, 7, 9\}$$

$$Y = \{8, 3, 7, 5, 6\}$$

means:-

$$\bar{X} = 6.2$$

$$\bar{Y} = 5.8$$

Products:-

$X_i - \bar{X}$	$Y_i - \bar{Y}$	Products
-2.2	2.2	-4.84
-1.2	-2.8	3.36
-0.2	1.2	-0.24
0.8	-0.8	-0.64
2.8	0.7	0.56
Total		-1.8

$$\text{Sample Covariance} = \frac{-1.8}{4} = -0.45$$

Negative Covariance indicates the Variables tend to move in opposite directions.

⑥ Major steps in Data preprocessing

① Data cleaning

② Data Integration

③ Data Transformation

④ Data Reduction

⑤ Data Discretization

Preprocessing pipeline

Raw Data

Handle missing values.

Remove duplicate Records.

Standardize formats.

Detect & Remove outliers.

Normalize Data.

Discretize

Clean Dataset.

⑤ ④ Equal-Frequency (Equal-Depth) Binning

Data (27 Values): 13, 15, 16, 16, 19, 20, 20, 21, 22, 22, 25, 25, 25, 25, 30, 33, 33, 35, 35, 35, 35, 36, 40, 45, 46, 52, 70

Bin 1: 13, 15, 16, 16, 19, 20, 20, 21, 22

$$\text{mean} = \frac{162}{9} = 18$$

Bin 2: 22, 25, 25, 25, 25, 30, 33, 33, 35

$$\text{mean} = \frac{253}{9} = 28.11$$

Discretization

Definition:- Data discretization converts continuous values into intervals.

Importance:-

- * Reduces noise
- * Simplifies data
- * Improves classification
- * Speedup mining algorithms.
- * Makes patterns easier to interpret.

Equal width Binning:-

Range divided into equal sized intervals.

Ex:-

Data: 10, 20, 30, 40, 50

